METHODS

Study Design

- Cross-sectional multi-omics integration (Proteomics + Metabolomics)
- Sample size: N = 6064 (Filtered for Age 40-70)
- Dataset: LabTools HPP 10K

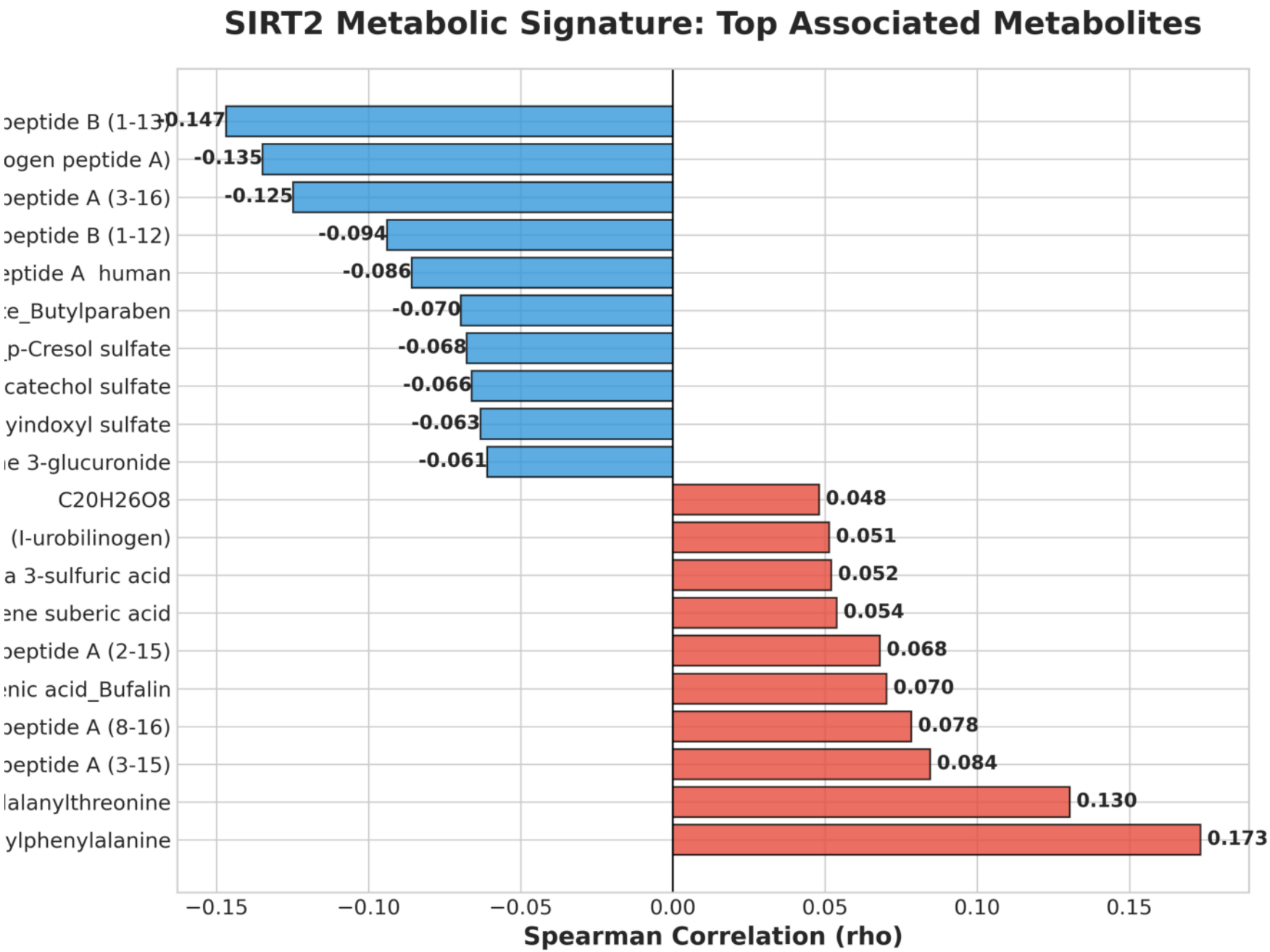
Statistical Analysis

- Model: Spearman Rank Correlation (non-parametric)
- Feature Selection: Top 10 positive and negative correlations for SIRT2
- Interpretation: rho > 0 (Positive Association), rho < 0 (Negative Association)

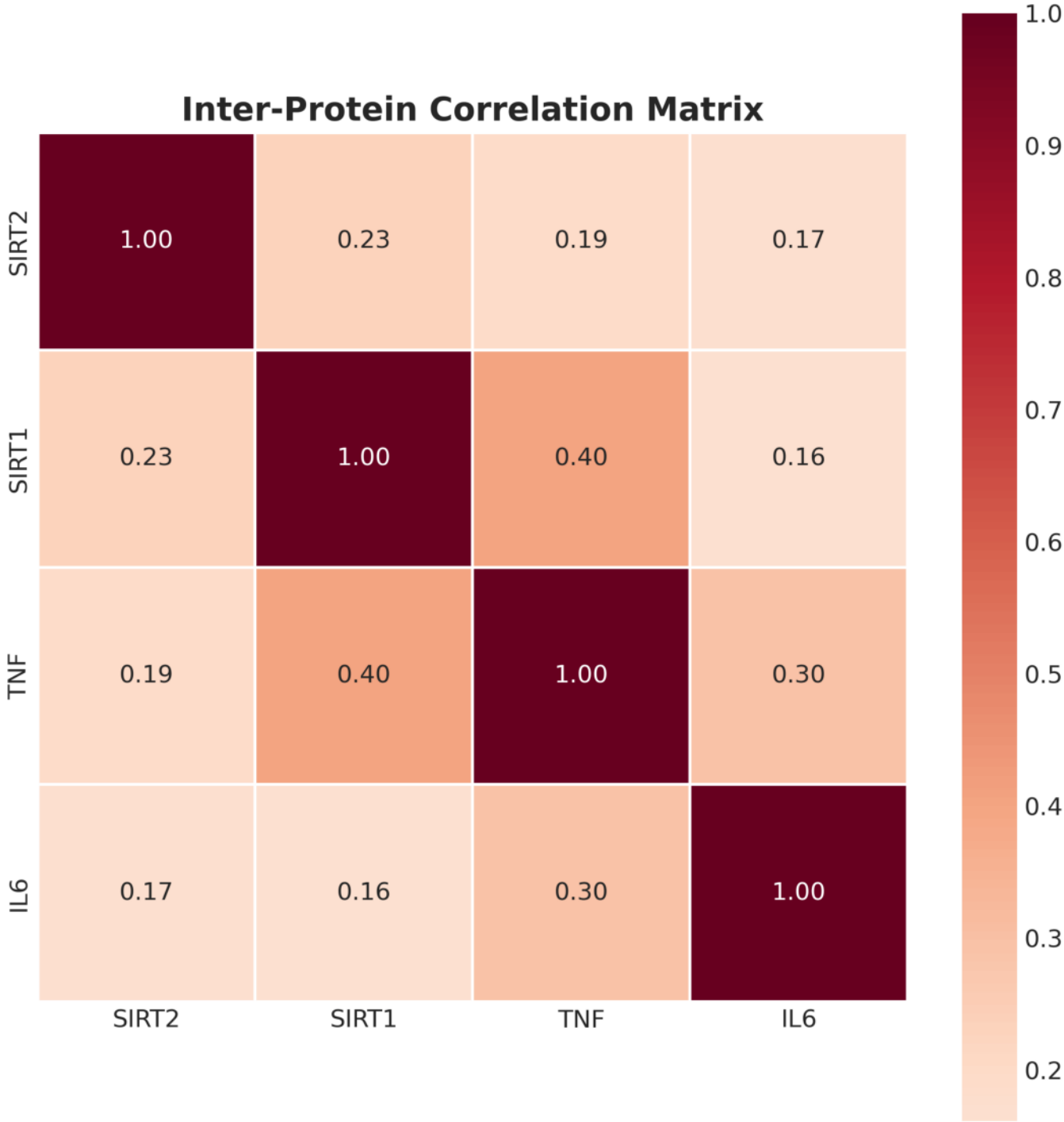
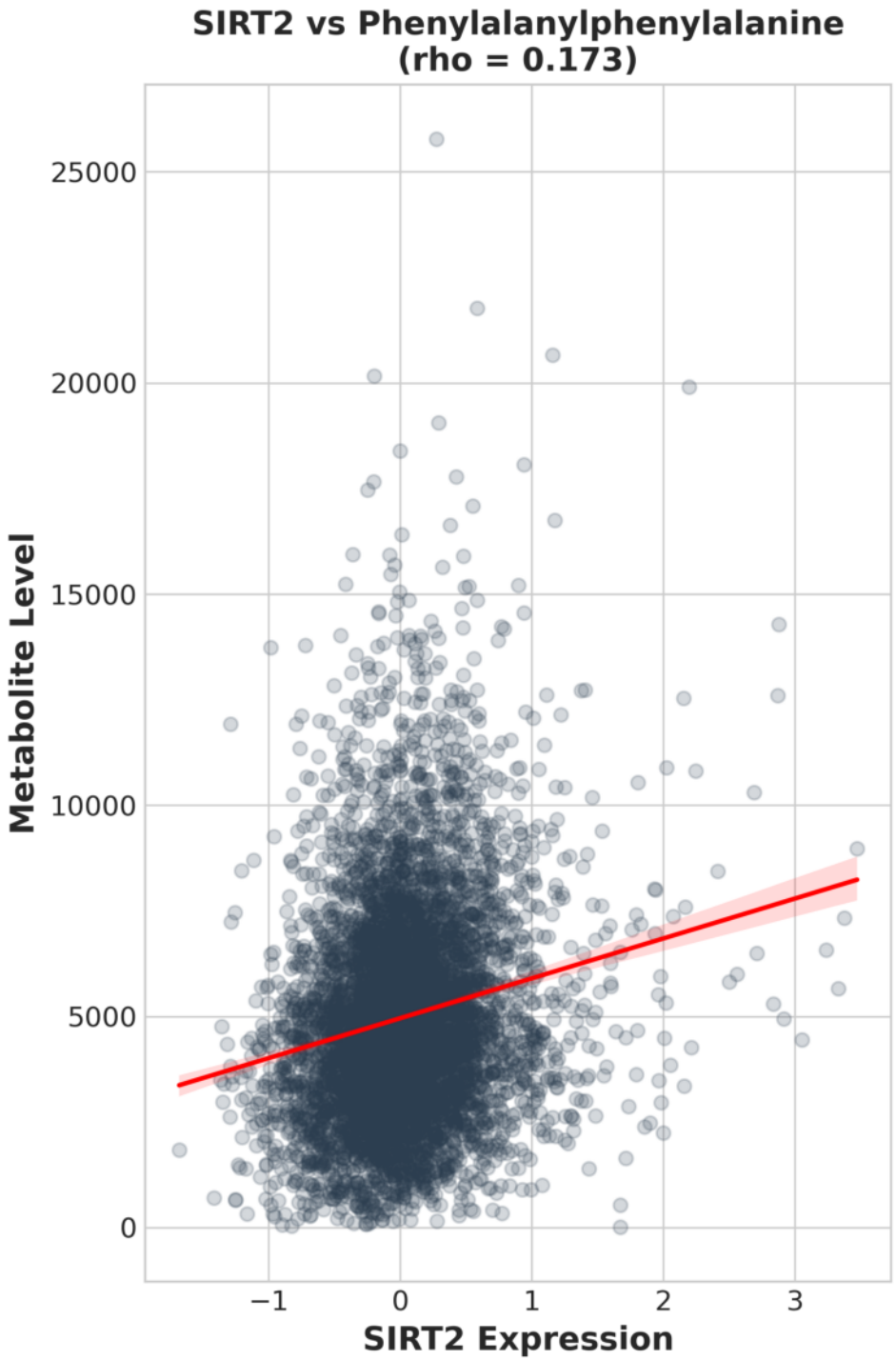
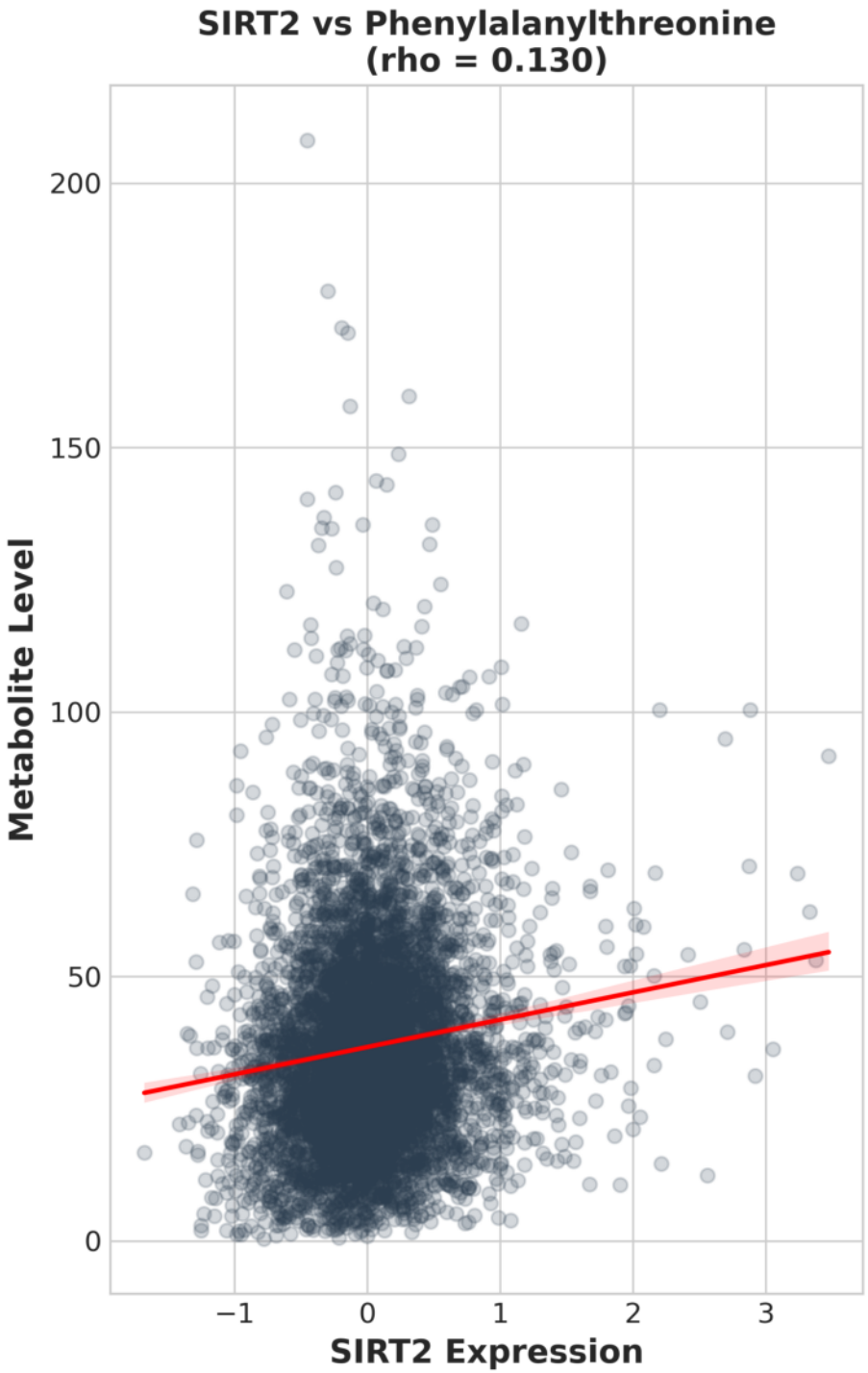
Inclusion Criteria

- Subjects aged 40-70 years
- Complete case analysis (dropna) for SIRT2 and associated metabolites

SIRT2 Metabolic Fingerprint



SIRT2 vs Inflammation Markers



Detailed Protein-Metabolite Coupling